

Explainable AIGC and Filter Detection on Edge Devices

Project Proposal

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Motivation

550%

Deepfake videos increase
2019 → 2024

Source: Deloitte 2024

10x

Deepfake incidents surge
2022 → 2023

Source: Sumsb 2023

1 in 5

American adults have used
a beauty filter

Source: Consumer Reports 2021

AI Image Generation & Filter Photorealistic AI faces and filtered photos flood social media — 95,000+ deepfake videos existed online by 2023. (Deloitte)

Social Impact In Feb 2024, a finance worker was tricked into wiring \$25M via deepfake video call (Arup case). CEO fraud targets 400+ companies per day. (Keepnet 2025)

Current Limitations Existing detectors output only Real/Fake with no explanation. Humans correctly identify high-quality deepfakes only 24.5% of the time. (MIT Media Lab)

Previous Work

Category I: Lightweight Detection

Representative Projects

Pixelproof, LFFD (arXiv 2019), DAD-HCNN (CVPRW 2020), RetouchingFFHQ (ACM MM 2023)

Methods & Results

- LFFD: Anchor-free CNN with receptive field chain. Optimized for edge devices — fast inference, low memory.
- DAD-HCNN: Hierarchical CNN distinguishing GAN-generated vs. retouched images. Two-stage: real/fake → GAN type. No public accuracy benchmark reported.
- RetouchingFFHQ (ACM MM 2023): First large-scale filter detection dataset (500K+ images). CNN baseline for beauty filter detection (smoothing, lifting, whitening).

Current Capabilities

Fast inference, suitable for initial screening. Verified: MobileNetV4 = 15.66ms/image, <1GB VRAM.

Fatal Limitation: The "Black-Box" Problem

Zero explainability. Binary output only. Users cannot know why a photo was flagged.

Previous Work

Category II: High Explainability

Representative Projects

FakeShield (ICLR 2025), FakeVLM (NeurIPS 2025)

Methods & Results

- FakeShield (ICLR 2025): MLLM + forgery region mask. MMTD-Set: 0.93 ACC on DeepFake, 0.98 ACC on AIGC-Editing. Outperforms MVSS-Net by +0.08 ACC on IMD2020. Explanation CSS score: 0.8873.
- FakeVLM (NeurIPS 2025): VLM fine-tuned on FakeClue (100K+ images). Evaluated on FakeClue, LOKI, DD-VQA. Matches expert binary classifiers + produces natural language artifact explanations. Open-source.

Current Capabilities

Outputs precise natural language reports + highlights specific artifact locations. State-of-the-art explainability.

Fatal Limitation: The Hardware Bottleneck

Verified FakeVLM: 7.24GB VRAM, ~10 seconds/image on T4 GPU. Requires cloud servers — cannot deploy to mobile.

Baseline Comparison

Why neither existing approach is sufficient

Criterion	Baseline 1 MobileNetV4	Baseline 2 FakeVLM	Our Goal
Inference Speed	15.66 ms ✓	~10,000 ms ✗	< 100 ms ✓
VRAM Required	< 1 GB ✓	7.24 GB ✗	< 8 GB ✓
Explainability	✗ None	✓ Full NL report	✓ Structured
Edge Deployment	✓ Mobile-ready	✗ Cloud only	✓ Mobile-ready
Filter Detection	✗ Not supported	✗ Not supported	✓ Supported
Open-source Code	✓ timm library	✓ GitHub	✓ Our repo

Gap: No existing method achieves both edge deployment AND explainability. This is our contribution.

Our goal

Explainable AIGC and Filter Detection on Edge Devices

Detect AI-generated images , Filter-processed images AND explain WHY



Building a deployable and lightweight backbone using MobileNetV4

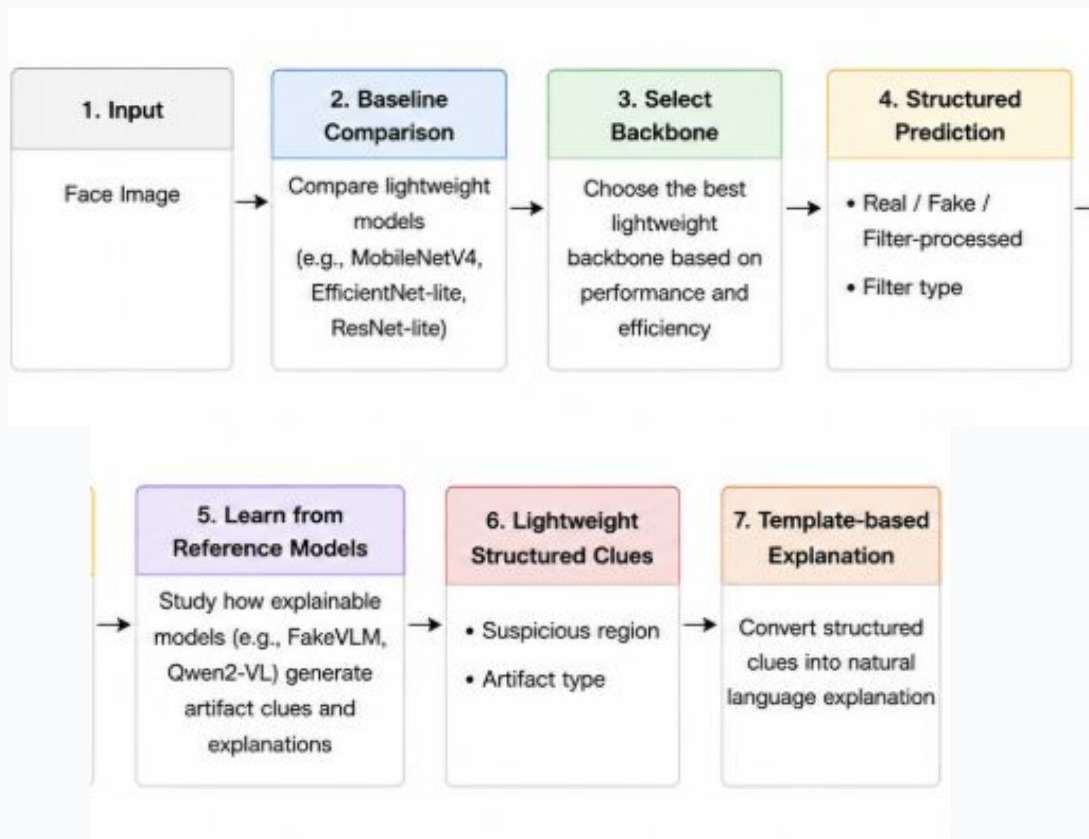


Output: Real/Fake/Filter-processed + Natural Language Explanation



Deploy as mobile app for real-world usage

Workflow Pipeline



Dataset

Data Sources

AlGuard_realfake_Dataset (real + fake)

Why: AIGC-specific dataset covering GAN and diffusion-generated faces. Needed to train the model on the latest AI-generation methods beyond traditional deepfakes.

FakeClue (real + fake)

Why: FakeVLM's official dataset. 100K+ images with fine-grained artifact annotations in natural language. Essential for explainability training — the only dataset providing artifact-level labels.

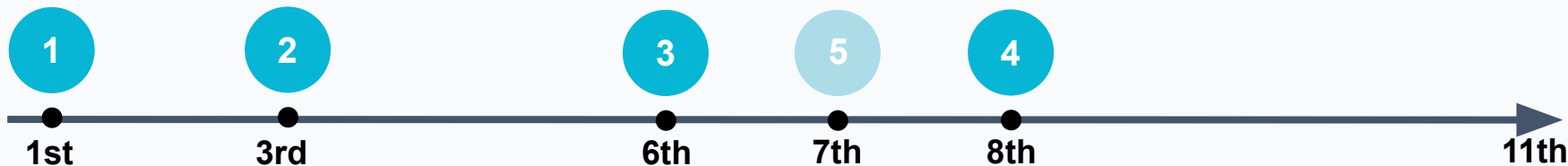
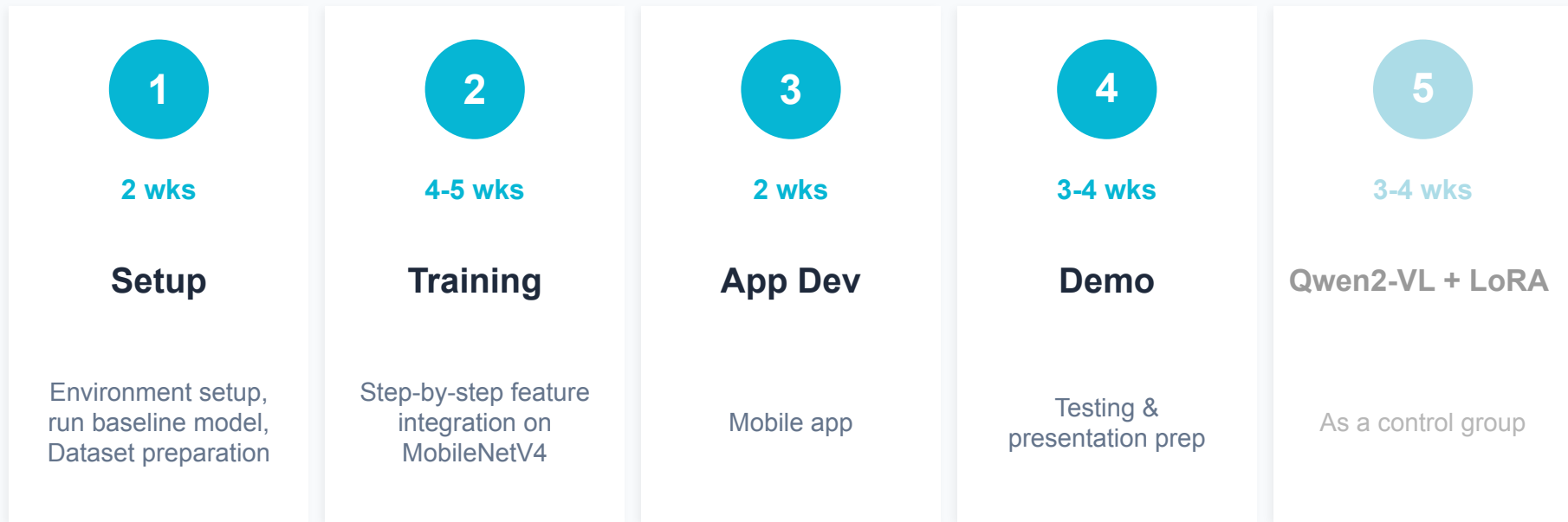
CelebA (real)

Why: Large-scale real face baseline with 200K+ celebrity face images. Provides clean real-face distribution; widely used and well-validated in the research community.

RetouchingFFHQ (real + filter)

Why: First large-scale filter detection dataset (ACM MM 2023). 500K+ fine-grained retouched images covering smoothing, lifting, whitening. Required for filter branch training — no other dataset covers beauty filter detection at this scale.

Schedule (11 weeks)



Computer resource

MobileNetV4

- GPU : NVIDIA RTX 3060/4060
- VRAM : 4 ~ 8GB
- RAM : 16GB
- storage : 50GB ~ 100GB

Qwen2-VL

- GPU : NVIDIA RTX 3090 / 4090 / A5000 / A6000
- VRAM $\geq$ 24GB
- RAM $\geq$ 32GB
- storage : 300GB ~ 500GB

Thank You

Questions & Discussion